

Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

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June 24, 2026

Recommendation

Use the latest completed assessment year, **2031-2032**, to identify teacher, course, and section patterns that deserve review before the next cycle. The stakeholder metric is **BOY/EOY score gain**: end-of-year score minus beginning-of-year score for the same public-safe student record.

The decision system does not ask whether old sections should be managed retroactively. It asks whether the most recent actual growth was materially above or below an expected-growth baseline trained only on prior years.

The baseline selected for operations is **Growth ensemble balanced**. It predicts **score gain directly** using a Validation ensemble specification. The selected tuning choice is an equal-weight blend of gradient boosting, GAM, and degree-3 polynomial growth predictions. The latest-year expected-gain RMSE is **4.660** points and the latest-year MAE is **3.725** points.

The latest-year gain R-squared is **0.110**; EOY R-squared is **0.933** and is reported only as secondary context. EOY is easier to predict because BOY score mechanically explains much of final score. The workflow uses the direct-growth model to create a fair expected-growth baseline, then makes decisions from aggregate teacher, course, and section residuals with bootstrap uncertainty checks.

Slice	Priority or watch rows	Total reviewed
Teachers	1	5
Courses	3	9
Sections	8	24

Decision	Slice	Target	N	Gap	95% CI	q
Intervention	Section	S19	12	-2.87	-5.29 to -0.17	0.160
Intervention	Section	S21	12	-2.71	-5.10 to -0.36	0.160
Intervention	Teacher	TCH-005	68	-1.26	-2.43 to -0.23	0.100
Bright spot	Section	S13	9	+2.51	0.54 to 4.44	0.080
Bright spot	Section	S02	10	+2.51	0.09 to 4.95	0.160

The full decision table, including watch-list rows, is generated as [reports/intervention_targets.csv](#).

Plain-English Method

1. Build a paired BOY/EOY growth extract from public-safe assessment records.
2. Define the business outcome as score gain: EOY score minus BOY score.
3. Train candidate expected-growth models on prior years only, including parametric, smooth, tree-based, ensemble, and excluded leakage-check specifications.
4. Select the operating baseline using leave-one-year-out temporal validation, with repeated CV and bootstrap checks as supporting evidence.
5. Score the latest year against that prior-year baseline.
6. Aggregate observed-minus-expected growth by teacher, course, and section.
7. Flag review targets only when the gap is large enough to matter and the uncertainty check supports follow-up.

This design separates the prediction problem from the decision problem. The prediction model estimates what growth would be expected for a similar starting profile; the review layer asks where actual growth departed from that expectation.

Direct Answers

1. The analysis covers 1,737 paired BOY/EOY records across 174 section-year groups and 5 public-safe teacher identifiers.
2. The training window is 2025-2026, 2026-2027, 2027-2028, 2028-2029, 2029-2030, 2030-2031; the action year is 2031-2032.
3. The average raw gain across the full extract is 5.72 points; the latest-year raw gain is 5.34 points.
4. The model search tested 19 candidate baselines across parametric, nonlinear, ensemble, and leakage-check families.
5. The selected direct-growth baseline has temporal expected-gain RMSE 4.631, temporal MAE 3.711, and latest-year expected-gain RMSE 4.660.
6. Teacher, course, and section flags are audit priorities. They are not automatic personnel ratings or causal claims.

Data Audit

A record enters the growth model only when the same public-safe student has valid BOY and EOY scores in the same section and teacher context. This keeps improvement tied to one instructional experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique simulated teachers	5
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

Model Selection

The model search used 1,485 prior-year pairs and held out 252 latest-year pairs for action-year evaluation. The primary selection metric is leave-one-year-out temporal expected-gain RMSE, not latest-year performance, so the system does not choose a model by looking at the year it later reviews.

The best eligible direct-growth temporal RMSE was **4.630** from **Growth ensemble nonlinear weighted**. The selected model’s temporal RMSE was **4.631**, a difference of 0.001 points. Because that difference is below the 0.01-point practical tolerance, the selected model is the operating baseline because it has the strongest repeated-CV RMSE among the temporally tied direct-growth candidates.

The model-search guardrails were:

- Use direct BOY/EOY score gain as the operating target because that is the stakeholder performance metric.
- Select by temporal-CV RMSE so the baseline is judged on year-to-year generalization.
- Use repeated-CV RMSE as the tie-breaker when temporal-CV RMSE differs by less than 0.01 points.
- Keep teacher, course, and section identifiers out of the operating baseline because those are the review slices.
- Report EOY R-squared only as context because final score is mechanically easier to predict than growth.

Family	Candidates	Eligible	Best model	Selected	Temporal RMSE	Temporal R2	Latest RMSE
Validation ensemble	2	2	Growth ensemble balanced	Yes	4.631	0.172	4.660
Non-parametric boosting	2	2	GBM 1		4.661	0.161	4.803
Semi-parametric GAM	2	2	GAM k4		4.665	0.160	4.640
Parametric polynomial	2	2	Growth poly d3		4.666	0.160	4.591
EOY-derived benchmark	2	0	EOY readiness		4.683	0.153	4.626
Parametric linear	2	2	Growth readiness		4.683	0.153	4.626
Leakage benchmark	1	0	Leakage check		4.706	0.145	4.658
Parametric interactions	1	1	Growth in-teractions		4.719	0.140	4.614

Family	Candidates	Eligible	Best model	Selected	Temporal RMSE	Temporal R2	Latest RMSE
Non-parametric random forest	2	2	RF 1		4.729	0.136	4.805
Non-parametric tree	2	2	Tree 1		4.730	0.135	4.859
Parametric cyclic	1	1	Growth cyclic		4.744	0.132	4.617

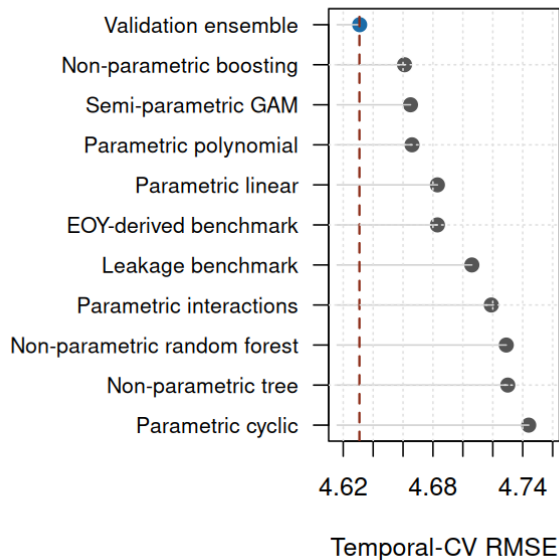
Family	Representative model	Decision	Temporal RMSE	Latest RMSE
Direct growth linear baselines	Growth readiness	Compared	4.683	4.626
Direct growth polynomial terms	Growth poly d3	Compared	4.666	4.591
Direct growth interaction surfaces	Growth interactions	Compared	4.719	4.614
GAM smooths	GAM k4	Compared	4.665	4.640
Regression trees	Tree 1	Compared	4.730	4.859
Random forests	RF 1	Compared	4.729	4.805
Gradient boosting	GBM 1	Compared	4.661	4.803
Validation ensembles	Growth ensemble balanced	Selected	4.631	4.660
EOY-derived benchmarks	EOY readiness	Compared	4.683	4.626
Teacher/course ID leakage check	Leakage check	Excluded	4.706	4.658

Model	Use	Family	CV RMSE	Temporal RMSE	Temporal R2	Latest RMSE	Latest R2
Growth ensemble nonlinear weighted		Validation ensemble	4.646	4.630	0.172	4.690	0.098
Growth ensemble balanced	Yes	Validation ensemble	4.640	4.631	0.172	4.660	0.110
GBM 1		Non-parametric boosting	4.692	4.661	0.161	4.803	0.054

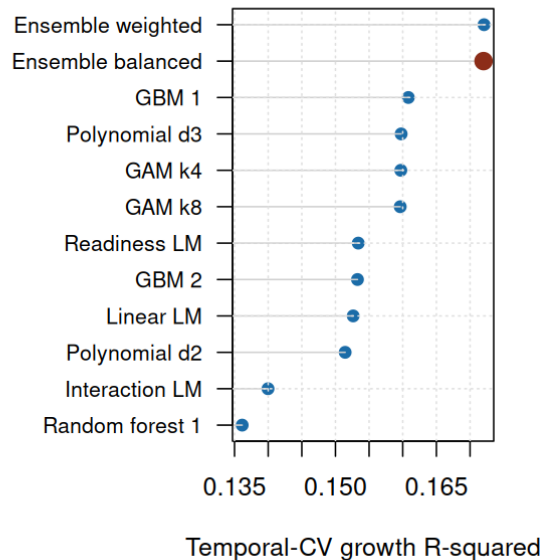
Model	Use	Family	CV RMSE	Temporal RMSE	Temporal R2	Latest RMSE	Latest R2
GAM k4		Semi-parametric GAM	4.660	4.665	0.160	4.640	0.118
GAM k8		Semi-parametric GAM	4.660	4.666	0.160	4.640	0.118
Growth poly d3		Parametric polynomial	4.642	4.666	0.160	4.591	0.136
GBM 2		Non-parametric boosting	4.722	4.682	0.153	4.813	0.050
Growth readiness		Parametric linear	4.650	4.683	0.153	4.626	0.123
Growth linear		Parametric linear	4.677	4.684	0.153	4.683	0.101
Growth poly d2		Parametric polynomial	4.648	4.689	0.151	4.627	0.122
Leakage check		Leakage benchmark	4.678	4.706	0.145	4.658	0.111

Full model artifacts: [reports/growth_model_comparison_display.csv](#), [reports/growth_model_search_grid.csv](#), [reports/growth_model_family_summary.csv](#), [reports/growth_model_selection_rationale.csv](#), [reports/model_temporal_validation.csv](#), and [reports/model_bootstrap_validation.csv](#).

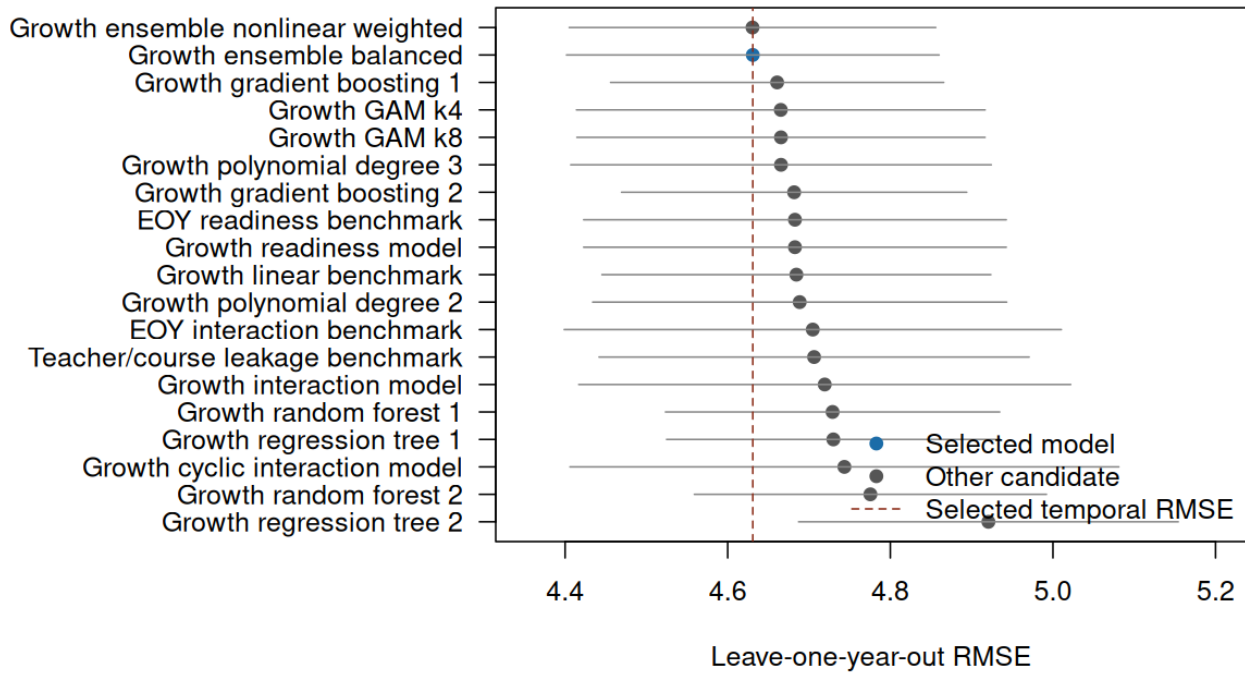
Best Candidate by Family



Top Growth Models



Expected-Growth Model Comparison



Latest-Year Review Targets

The latest-year review layer compares observed gain with expected gain for 24 section groups. The decision labels use a practical audit threshold: material gap, bootstrap interval direction, and BH-adjusted q-value for multiple-review control.

Teacher review

Target	N	Raw	Expected	Gap	CI	q	Decision
TCH-005	68	3.81	5.08	-1.26	-2.43 to -0.23	0.100	Intervention

Course review

Target	N	Raw	Expected	Gap	CI	q	Decision
Precalc	40	4.99	6.13	-1.13	-2.48 to 0.00	0.160	Watch
AP Calc AB	49	4.31	5.39	-1.08	-2.51 to 0.10	0.216	Watch
AP Precalc	30	6.11	4.80	+1.31	-0.30 to 2.59	0.216	Watch

Section evidence

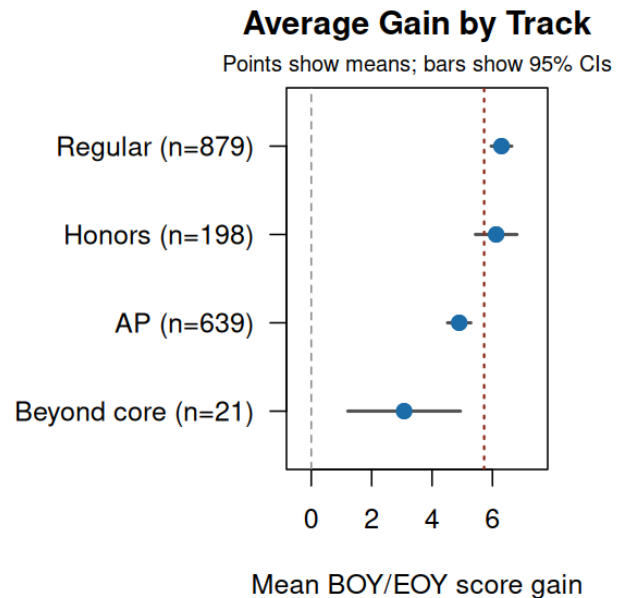
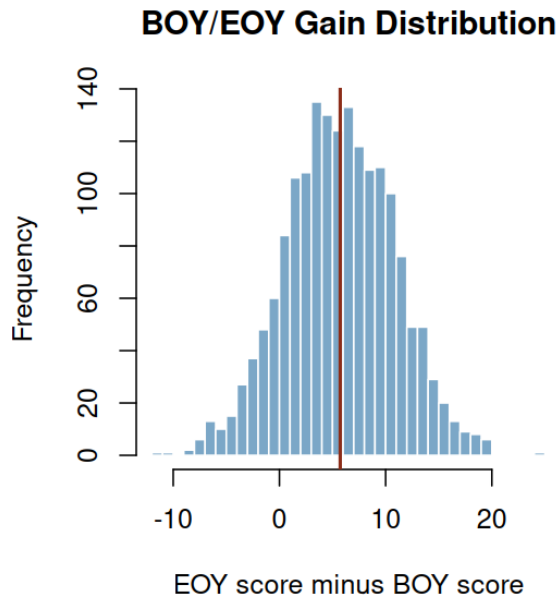
Target	N	Raw	Expected	Gap	CI	q	Decision
S19	12	2.41	5.28	-2.87	-5.29 to -0.17	0.160	Intervention
S21	12	3.10	5.81	-2.71	-5.10 to -0.36	0.160	Intervention
S13	9	7.18	4.66	+2.51	0.54 to 4.44	0.080	Bright spot
S02	10	9.02	6.52	+2.51	0.09 to 4.95	0.160	Bright spot
S04	9	4.01	6.78	-2.76	-6.94 to 2.23	0.553	Watch
S23	9	1.89	4.14	-2.26	-4.80 to 0.25	0.293	Watch
S16	10	4.11	5.99	-1.87	-4.78 to 0.49	0.457	Watch
S05	11	4.00	5.28	-1.29	-3.93 to 1.74	0.651	Watch

Full review tables: [reports/latest_teacher_review.csv](#), [reports/latest_course_review.csv](#), and [reports/latest_section_review.csv](#).

Section Evidence

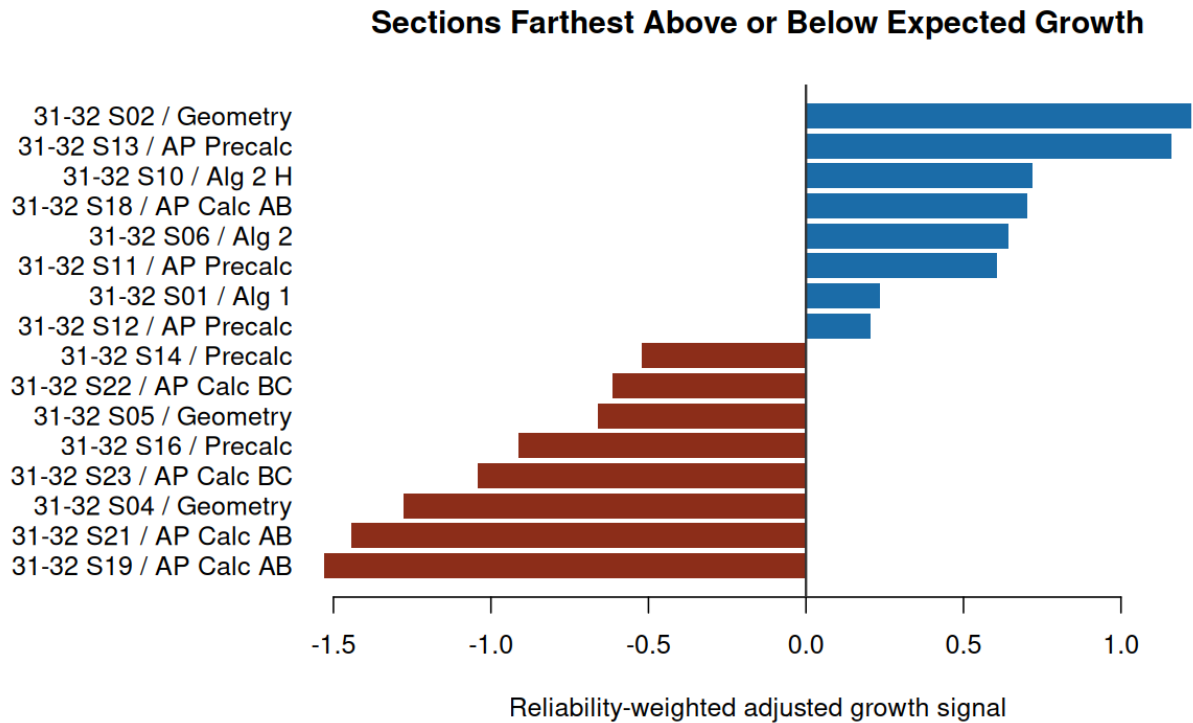
Raw section improvement is useful for communication, but it is not the final comparison. A section can show positive raw gain and still fall below expected growth if its starting profile suggested a larger increase.

Section	Course	N	BOY	EOY	Gain	95% CI	p-value
S02	Geometry	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
S13	AP Precalc	9	63.5	70.7	7.18	5.20 to 9.16	<0.001
S10	Alg 2 H	12	63.1	70.0	6.90	3.98 to 9.83	<0.001
S18	AP Calc AB	13	49.4	56.1	6.69	3.06 to 10.31	0.002
S06	Alg 2	11	41.0	48.6	7.63	4.97 to 10.30	<0.001
S11	AP Precalc	10	69.0	74.7	5.67	1.52 to 9.83	0.013
S01	Alg 1	13	42.1	49.2	7.13	4.19 to 10.08	<0.001
S12	AP Precalc	11	59.5	65.1	5.64	1.86 to 9.43	0.008



The adjusted section signal is observed gain minus expected gain, reliability-weighted toward zero for smaller sections.

Section	Teacher	Course	N	Raw	Expected	Signal	Result
S02	TCH-002	Geometry	10	9.02	6.52	+1.22	In range
S13	TCH-004	AP Precalc	9	7.18	4.66	+1.16	Above
S10	TCH-003	Alg 2 H	12	6.90	5.56	+0.72	In range
S18	TCH-005	AP Calc AB	13	6.69	5.41	+0.70	In range
S16	TCH-004	Precalc	10	4.11	5.99	-0.91	In range
S05	TCH-001	Geometry	11	4.00	5.28	-0.66	In range
S22	TCH-005	AP Calc BC	10	3.10	4.36	-0.61	In range
S14	TCH-003	Precalc	10	4.81	5.88	-0.52	In range



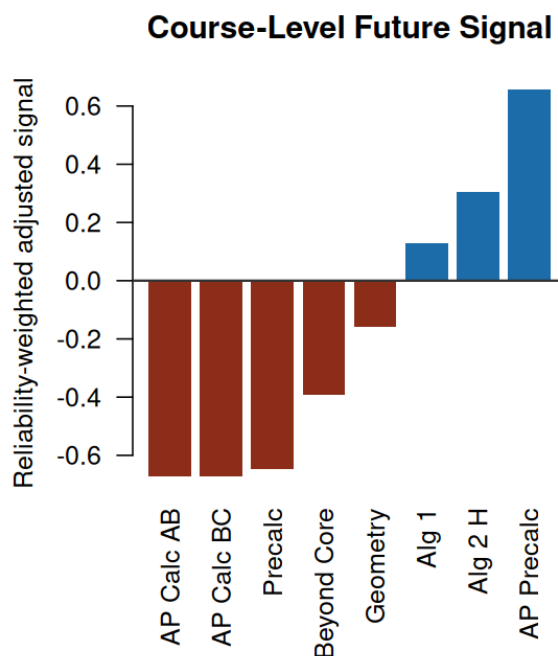
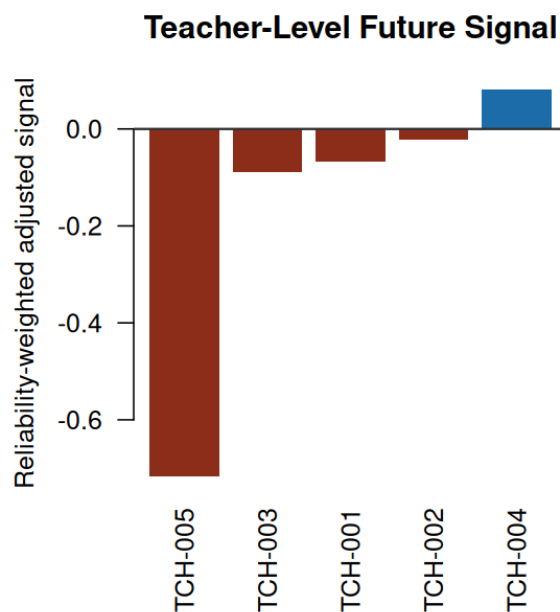
The full section signal table is generated as [reports/section_adjusted_signals.csv](#).

Teacher and Course Summaries

These summaries aggregate the latest-year evidence into planning views. They support review conversations about pacing, curriculum alignment, attendance mix, and transferable practices. They should not be read as standalone personnel scores.

Teacher	Sections	Records	Raw	Expected	Signal
TCH-004	5	43	5.21	5.03	+0.08
TCH-002	5	52	6.60	6.64	-0.02
TCH-001	3	33	6.06	6.23	-0.07
TCH-003	5	56	5.70	5.87	-0.09
TCH-005	6	68	3.81	5.08	-0.72

Course	Sections	Records	Raw	Expected	Signal
AP Precalc	3	30	6.11	4.80	+0.66
Alg 2 H	2	26	6.18	5.53	+0.30
Alg 1	1	13	7.13	6.71	+0.13
Alg 2	3	33	6.57	6.64	-0.04
Geometry	4	39	5.99	6.27	-0.16
Beyond Core	1	3	-0.11	4.21	-0.39
Precalc	4	40	4.99	6.13	-0.65
AP Calc BC	2	19	2.52	4.26	-0.67
AP Calc AB	4	49	4.31	5.39	-0.67



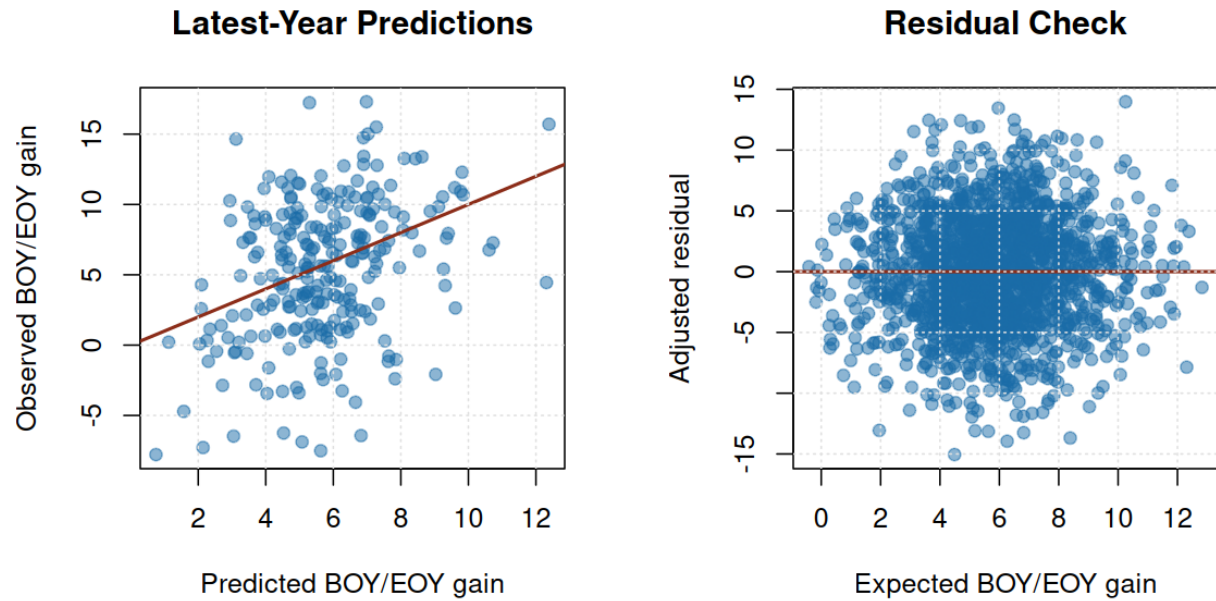
Diagnostics and Sensitivity

The baseline is strong for final-score expectation and weaker for individual gain variation. That pattern is expected: BOY score explains much of EOY score, while individual improvement contains more unobserved classroom, attendance, and assessment noise.

Diagnostic	Estimate	Interpretation
Latest-year expected-gain RMSE	4.660	Typical out-of-sample prediction error on latest-year BOY/EOY gain
Latest-year expected-gain R-squared	0.110	Share of latest-year gain variation explained by the expected-growth model
Latest-year EOY R-squared	0.933	Share of latest-year EOY score variation explained by the baseline
Latest-year residual mean	-0.380	Near 0 means expected gain is centered in the action year
Latest-year residual SD	4.654	Latest-year residual spread used for slice uncertainty

Metric	Estimate	95% interval
Expected-gain RMSE	4.660	4.226 to 5.081
Expected-gain MAE	3.725	3.359 to 4.071
Expected-gain R-squared	0.110	0.009 to 0.189
EOY RMSE	4.660	4.226 to 5.081
EOY R-squared	0.933	0.917 to 0.944

Measure	Value
Training paired records	1,485
Latest-year paired records	252
Latest-year section-year groups	24
Latest-year mean raw BOY/EOY gain	5.34
Latest-year raw-vs-adjusted rank correlation	0.897
Latest-year top-10 overlap, raw vs adjusted ranking	70.0%



Technical Appendix

The operating model excludes teacher IDs, course IDs, and section IDs. A leakage benchmark is reported only to show what would happen if persistent IDs were included in the baseline; it is not used for review because it would absorb the teacher/course patterns the decision layer is designed to detect.

Package	Installed
mgcv	Available
randomForest	Available
gbm	Available
rpart	Available

Metric	Value
Selected model	Growth ensemble balanced
Selected target strategy	Direct growth
Selected method	Ensemble
Selected family	Validation ensemble
Selected tuned parameters	Growth gradient boosting 1 weight=1; Growth GAM k4 weight=1; Growth polynomial degree 3 weight=1
Selection rule	Lowest repeated-CV RMSE among direct-growth candidates within 0.01 points of the best temporal-CV RMSE
Training paired records	1,485
Latest-year action paired records	252
Training years	2025-2026, 2026-2027, 2027-2028, 2028-2029, 2029-2030, 2030-2031

Metric	Value
Action year	2031-2032
Candidate models tested	19
Operational candidates tested	16
Excluded leakage benchmarks	3
Repeated CV folds	5
Repeated CV repeats	2
Temporal expected-gain RMSE	4.631
Temporal expected-gain MAE	3.711
Temporal expected-gain R-squared	0.172
Temporal expected-gain RMSE SD	0.229
Temporal EOY R-squared	0.938
Latest-year expected-gain RMSE	4.660
Latest-year expected-gain MAE	3.725
Latest-year expected-gain R-squared	0.110
Latest-year EOY R-squared	0.933

The full selected-model metrics table is generated as [reports/growth_final_metrics.csv](#).

Conclusion

The project should be read as a statistical decision-support system, not as a simple prediction demo. The strongest business value is the workflow: choose a validated expected-growth baseline, compare latest actual growth to that baseline, quantify uncertainty by slice, and translate the evidence into review priorities.

The recommended stakeholder action is to review the flagged teacher, course, and section patterns before the next assessment cycle. Priority targets deserve support or investigation; positive anomalies deserve study for transferable practices; watch-list rows deserve context review before escalation.

The important limitation is that the data are public-safe and generalized from an assessment workflow. The outputs demonstrate the analysis pattern and should not be used as real student, teacher, or personnel decisions.

Reproducibility

Rebuild the full evidence packet with `make all`. The pipeline uses the included public-safe extract and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.